UNIVERSIDAD ADOLFO IBÁÑEZ
FACULTAD DE INGENIERÍA Y CIENCIAS

Optimización

2026-2

Profesora: Dora Jiménez

SMART + SUSTAINABLE

FACULTAD DE INGENIERÍA Y CIENCIAS

UNIVERSIDAD ADOLFO IBÁÑEZ

Branch & Bound – Ramificación y Acotamiento

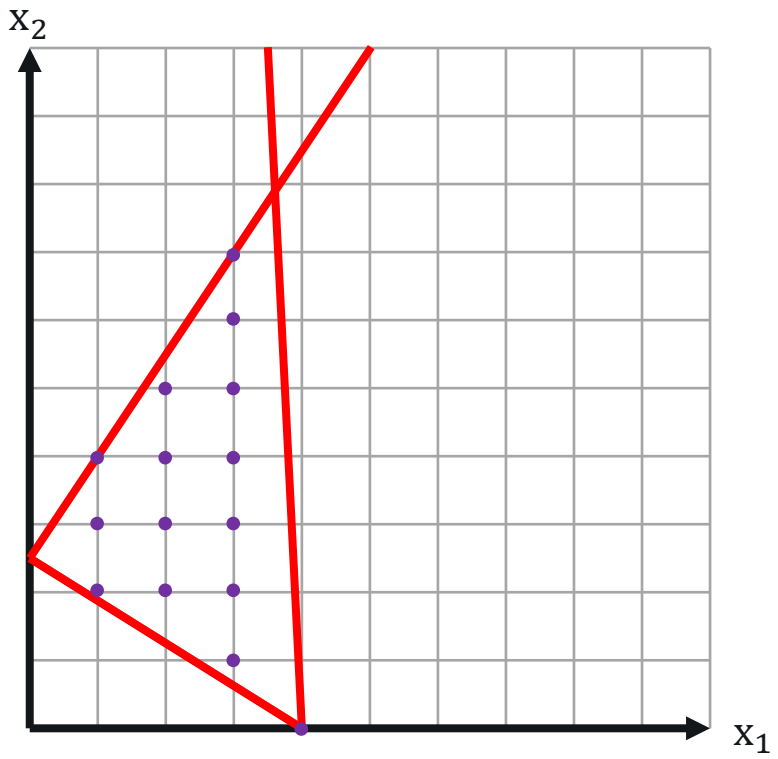
Resolvamos el siguiente problema utilizando el algoritmo de Branch & Bound

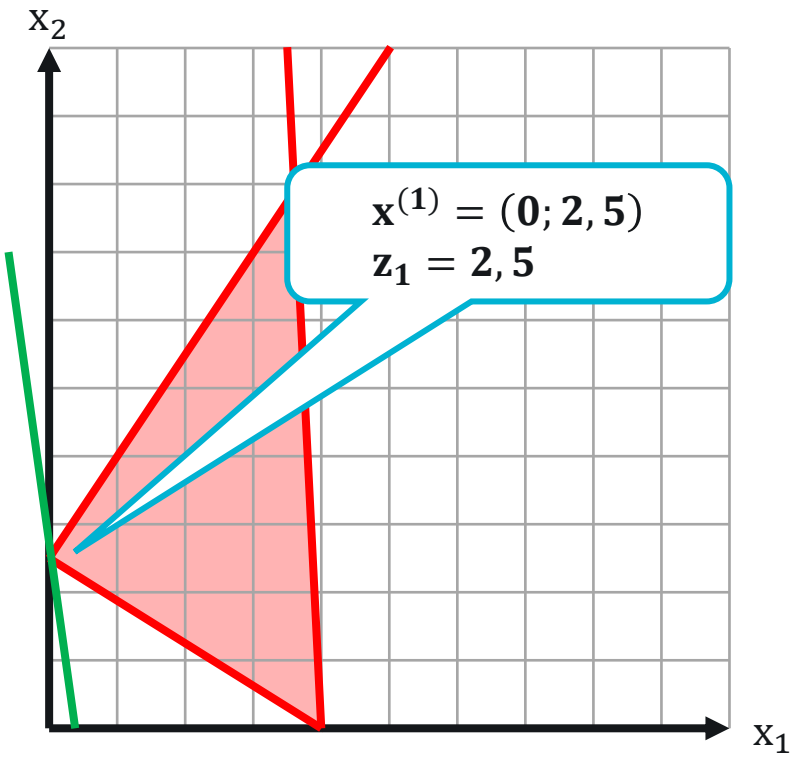
$$\text{MIN } Z = 7x_1 + x_2$$
$$\text{s. a.}$$
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_1, x_2 \in Z_0^+$$

Relajación lineal

$\mathbf{P_1}$

$$\text{MIN } Z = 7x_1 + x_2$$
$$\text{s. a.}$$
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_1, x_2 \geq 0$$

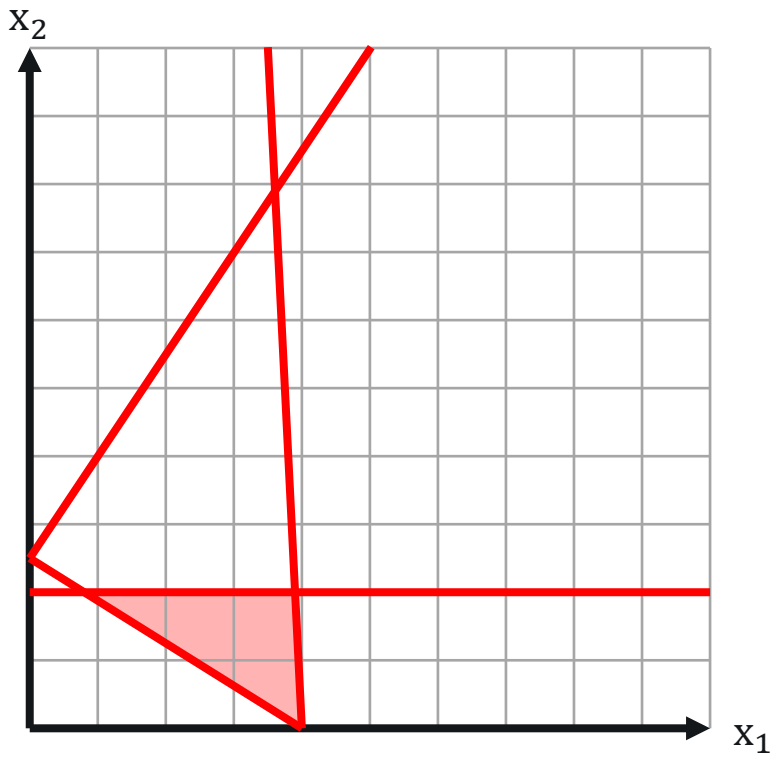




Por lo tanto, podemos ramificar P_1 en 2 sub problemas

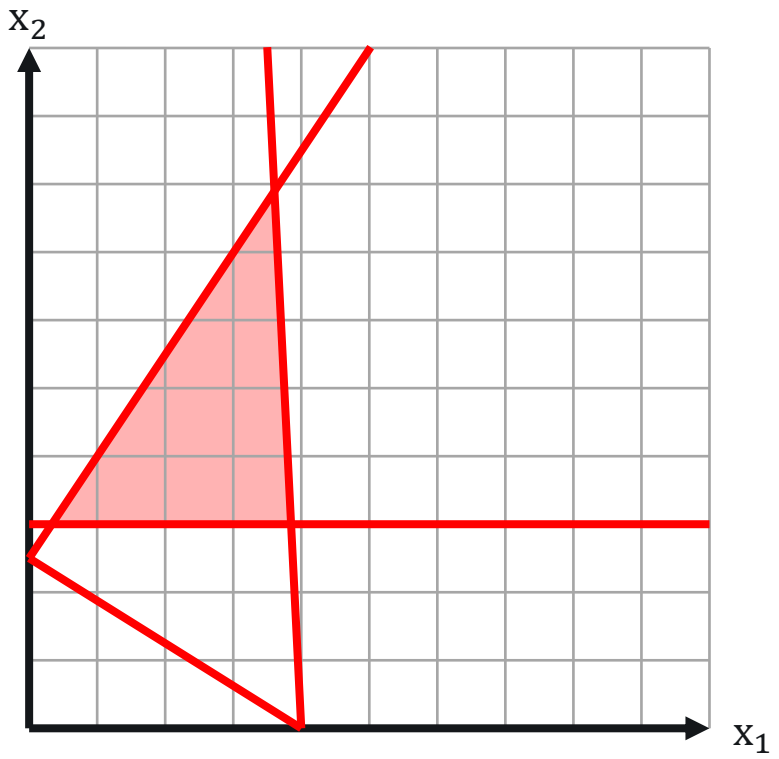
P_2

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2$$
$$x_1, x_2 \geq 0$$



P_3

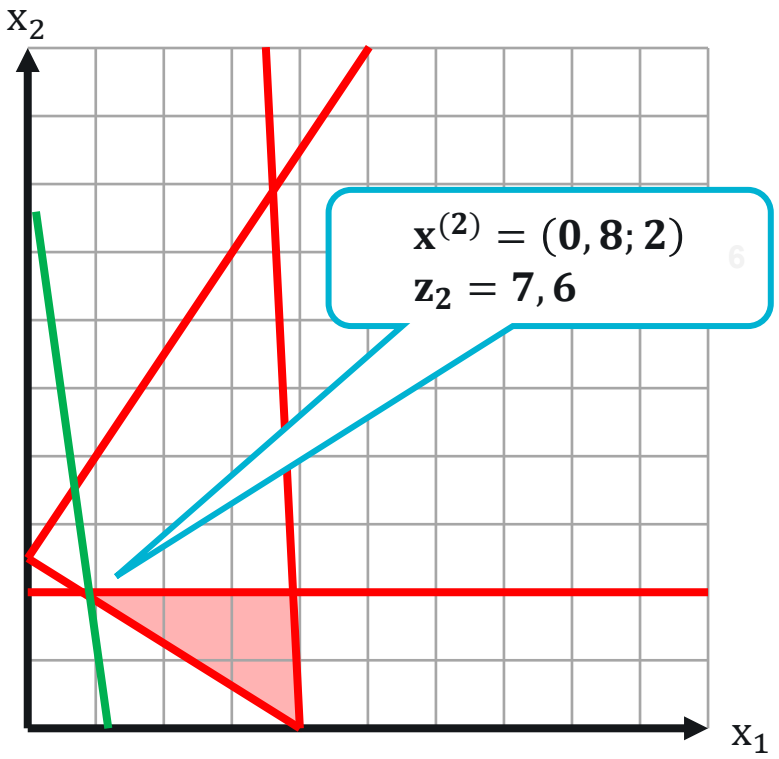
$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \geq 3$$
$$x_1, x_2 \geq 0$$

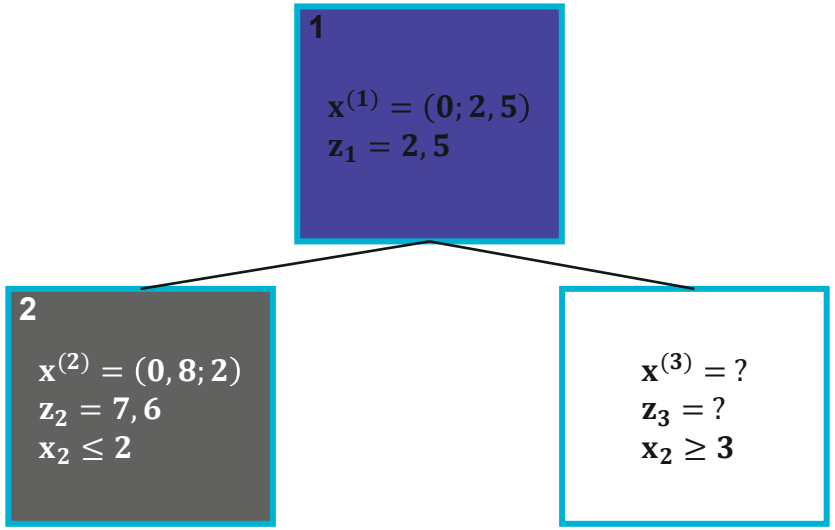


Procedamos entonces a resolver P_2

P_2

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2$$
$$x_1, x_2 \geq 0$$

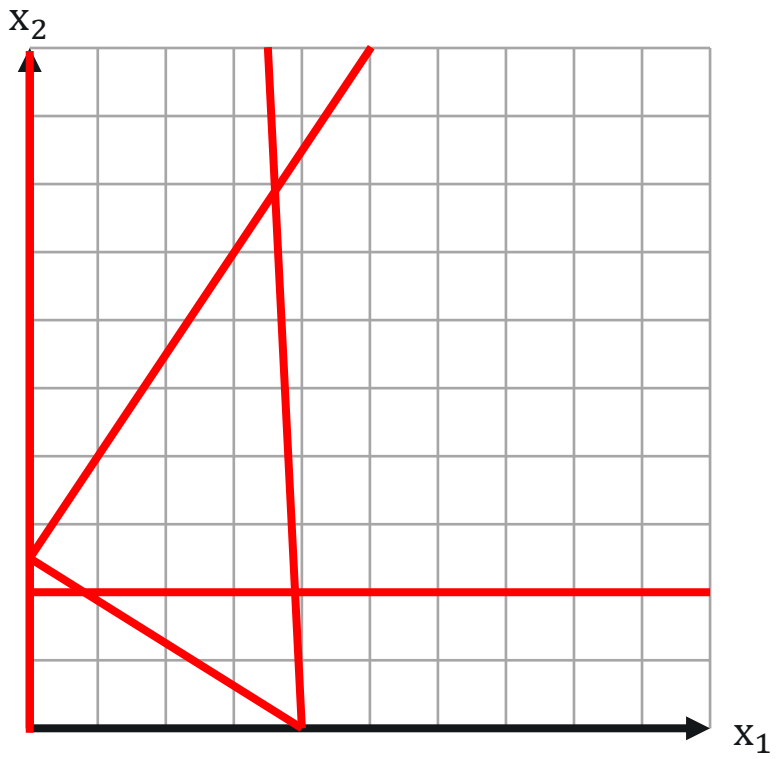




Por lo tanto, podemos ramificar P_2 en 2 sub problemas

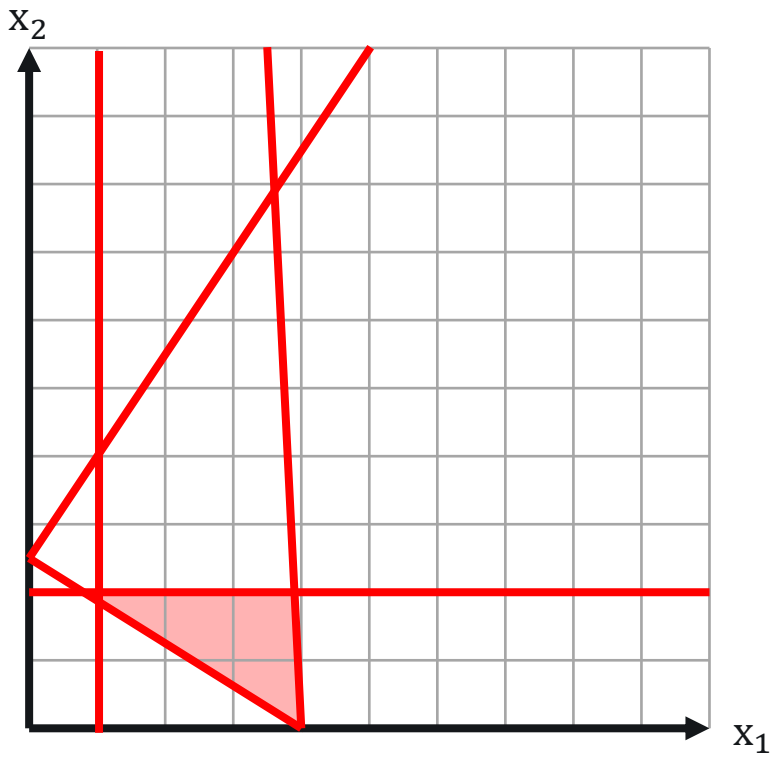
P_4

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2, x_1 \leq 0$$
$$x_1, x_2 \geq 0$$



P_5

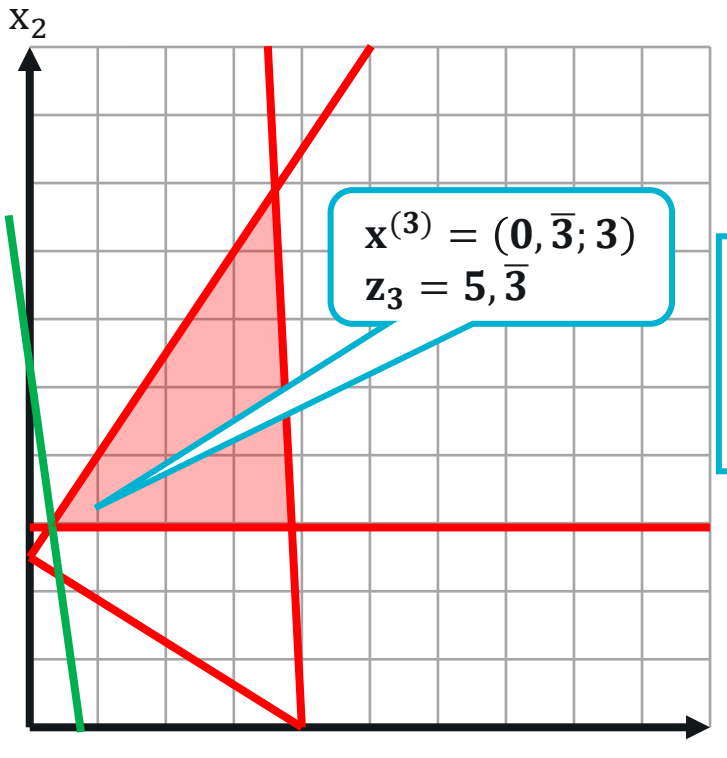
$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2, x_1 \geq 1$$
$$x_1, x_2 \geq 0$$

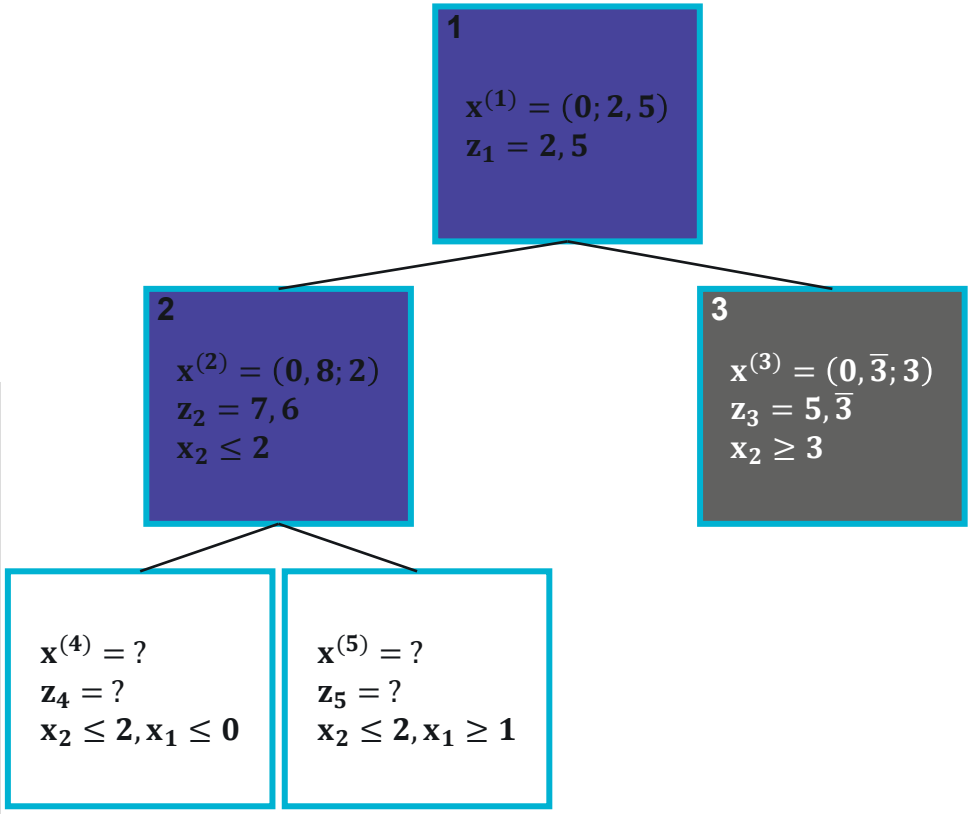


Procedamos entonces a resolver P_3

P_3

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \geq 3$$
$$x_1, x_2 \geq 0$$

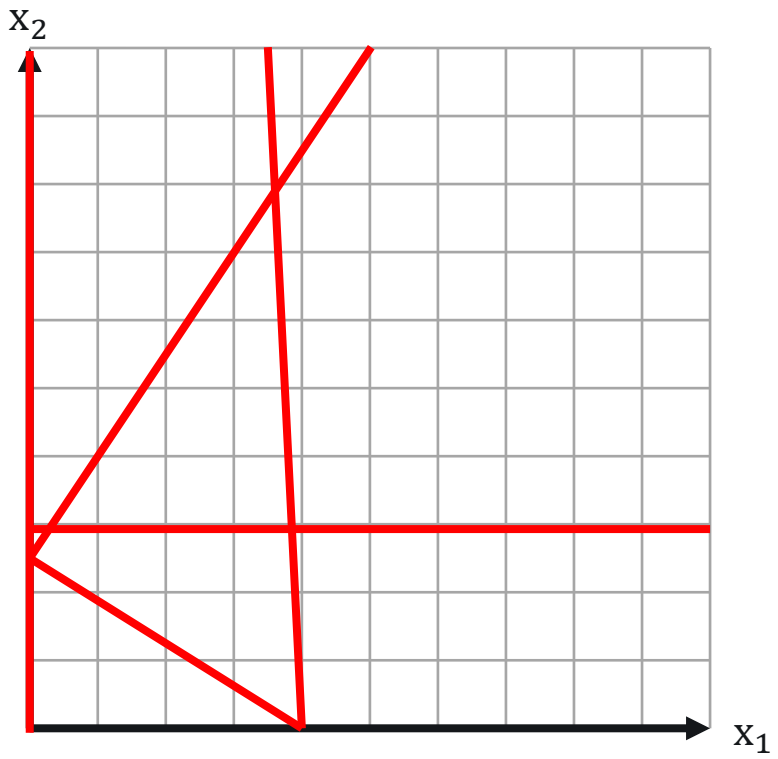




Por lo tanto, podemos ramificar P_3 en 2 sub problemas

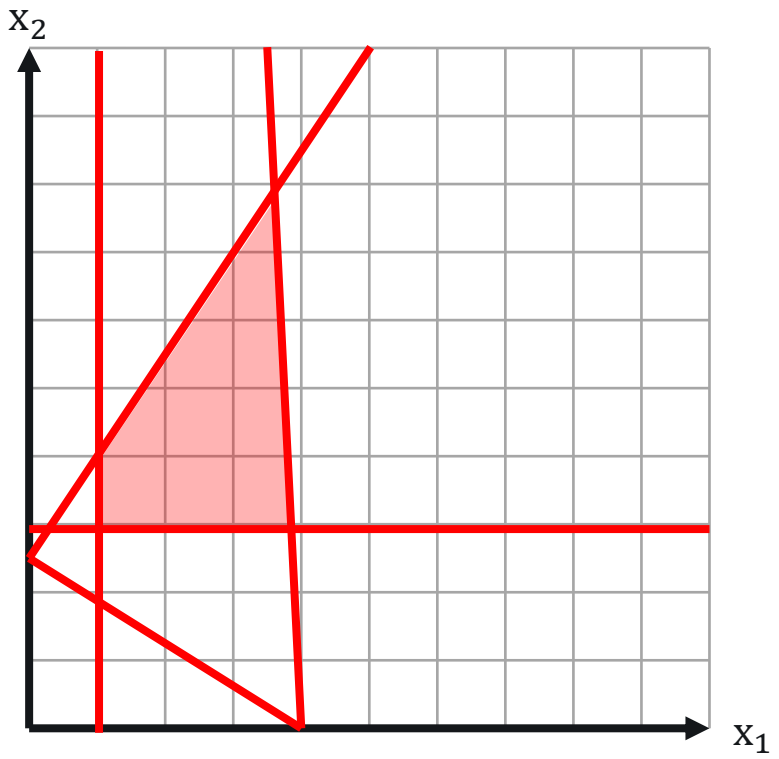
P_6

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \geq 3, x_1 \leq 0$$
$$x_1, x_2 \geq 0$$



P_7

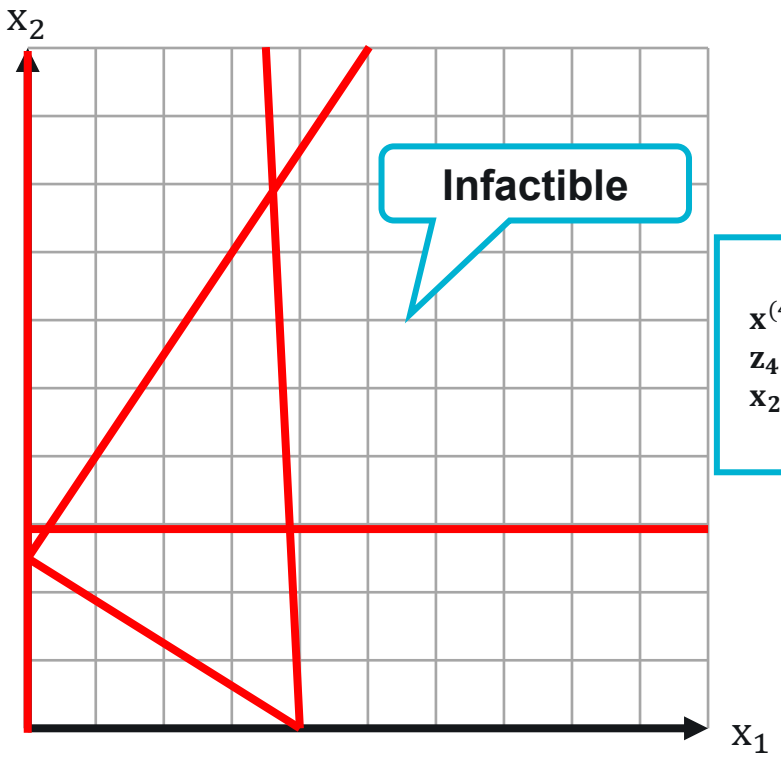
$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \geq 3, x_1 \geq 1$$
$$x_1, x_2 \geq 0$$

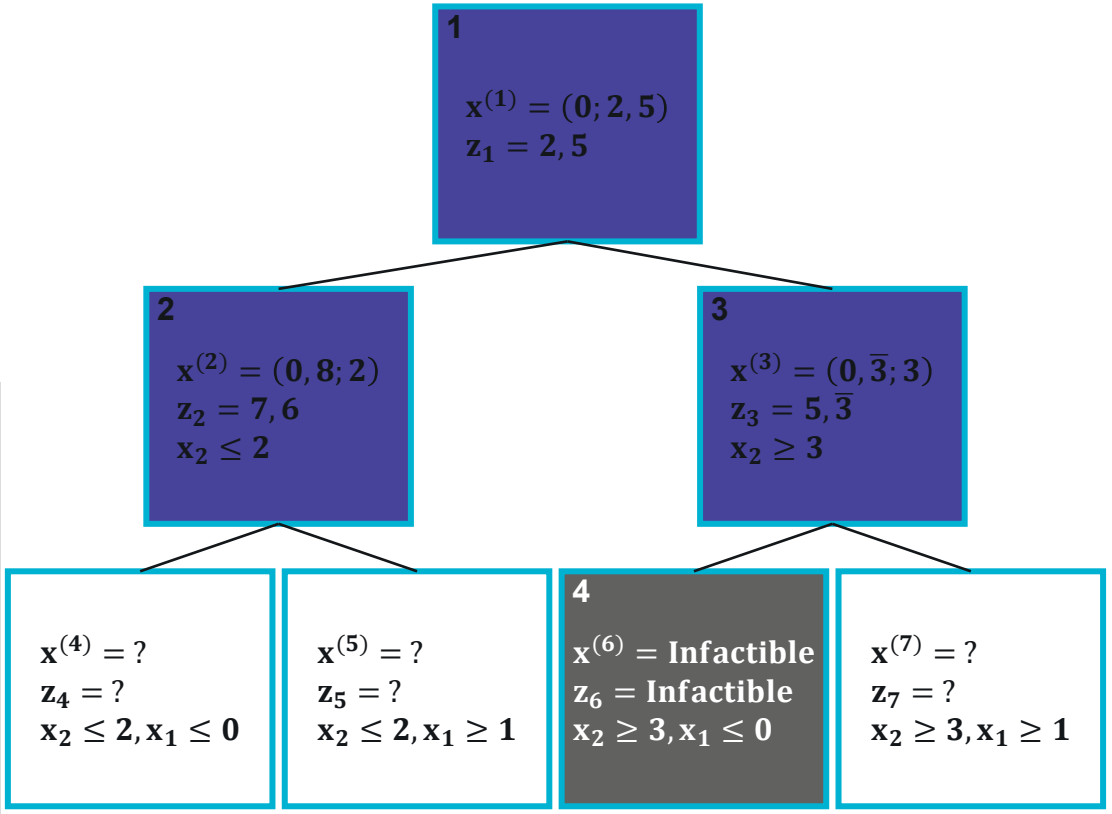


Procedamos entonces a resolver P_6, el cual nos damos cuenta que es infactible por lo que no lo ramificamos

P_6

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \geq 3, x_1 \leq 0$$
$$x_1, x_2 \geq 0$$

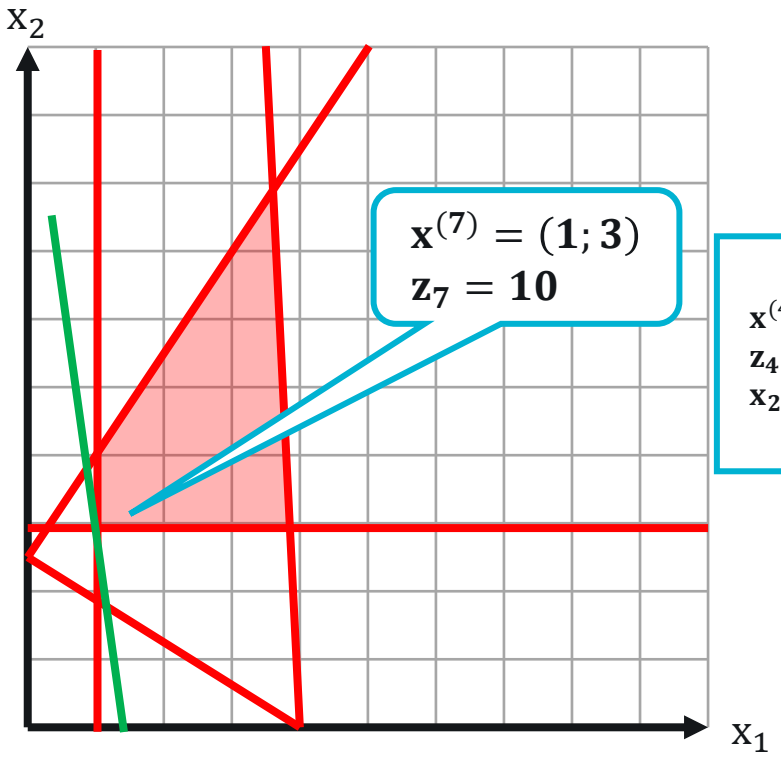


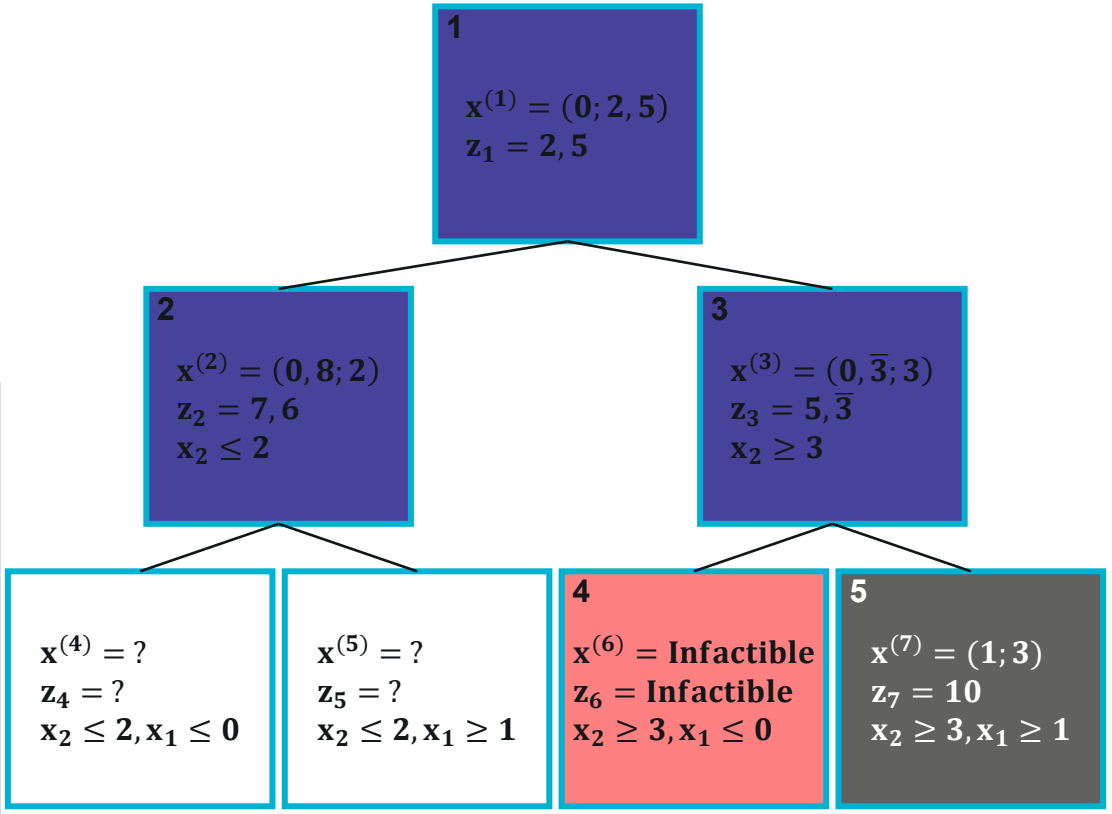


Procedamos entonces a resolver P_7, Cuyo óptimo es solución factible en el problema original, por lo que no ramificamos

P_7

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \geq 3, x_1 \geq 1$$
$$x_1, x_2 \geq 0$$

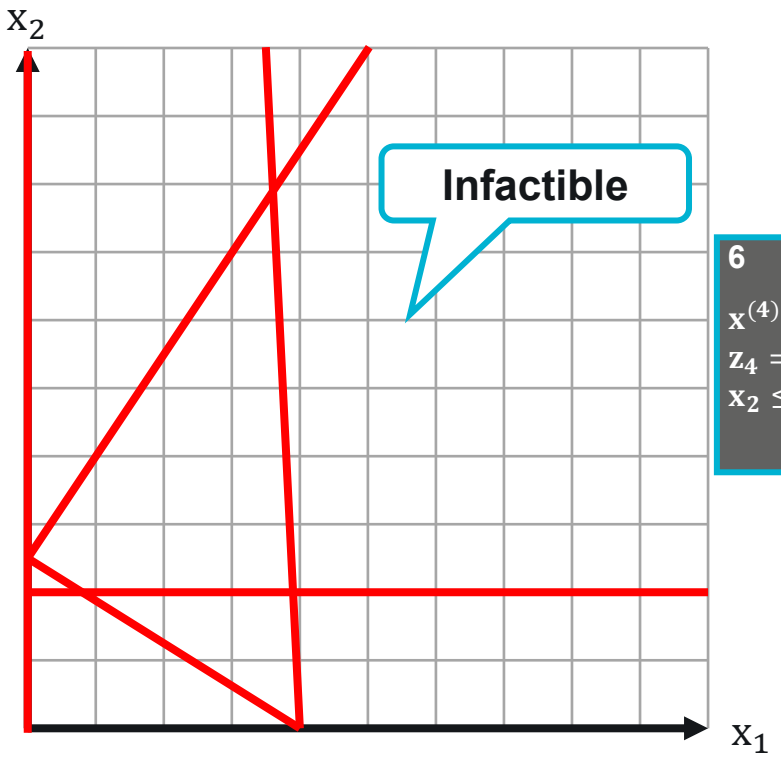


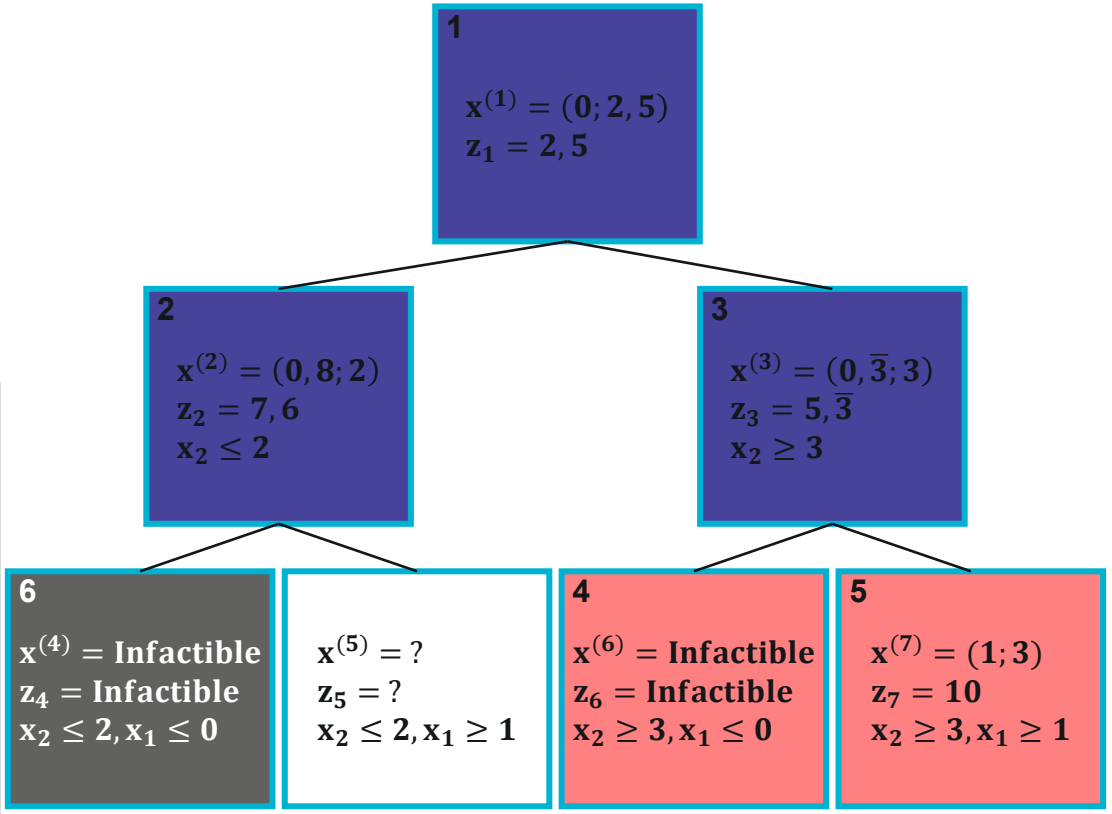


Procedamos entonces a resolver P_4, el cual nos damos cuenta que es infactible por lo que no lo ramificamos

P_4

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2, x_1 \leq 0$$
$$x_1, x_2 \geq 0$$

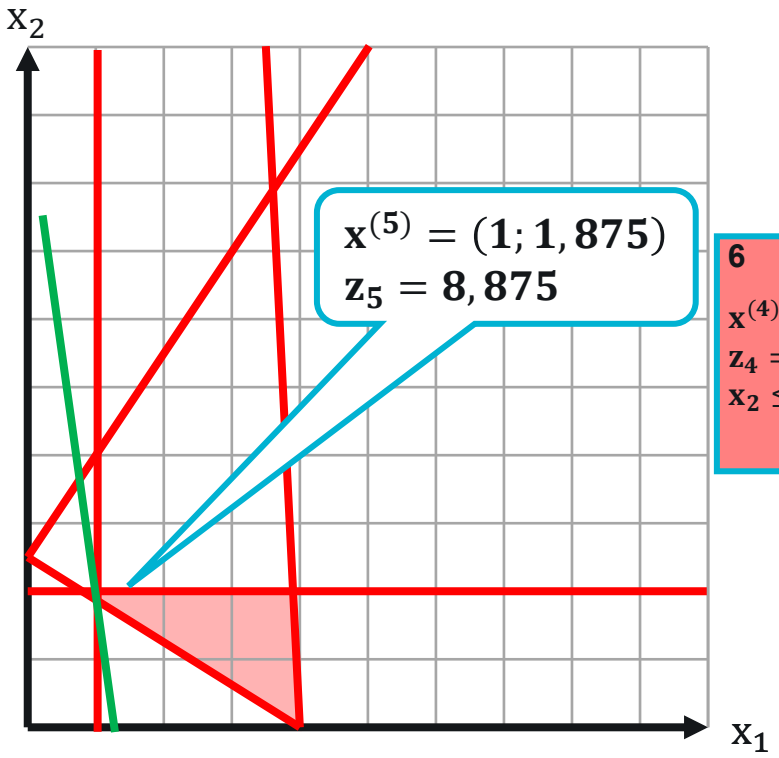


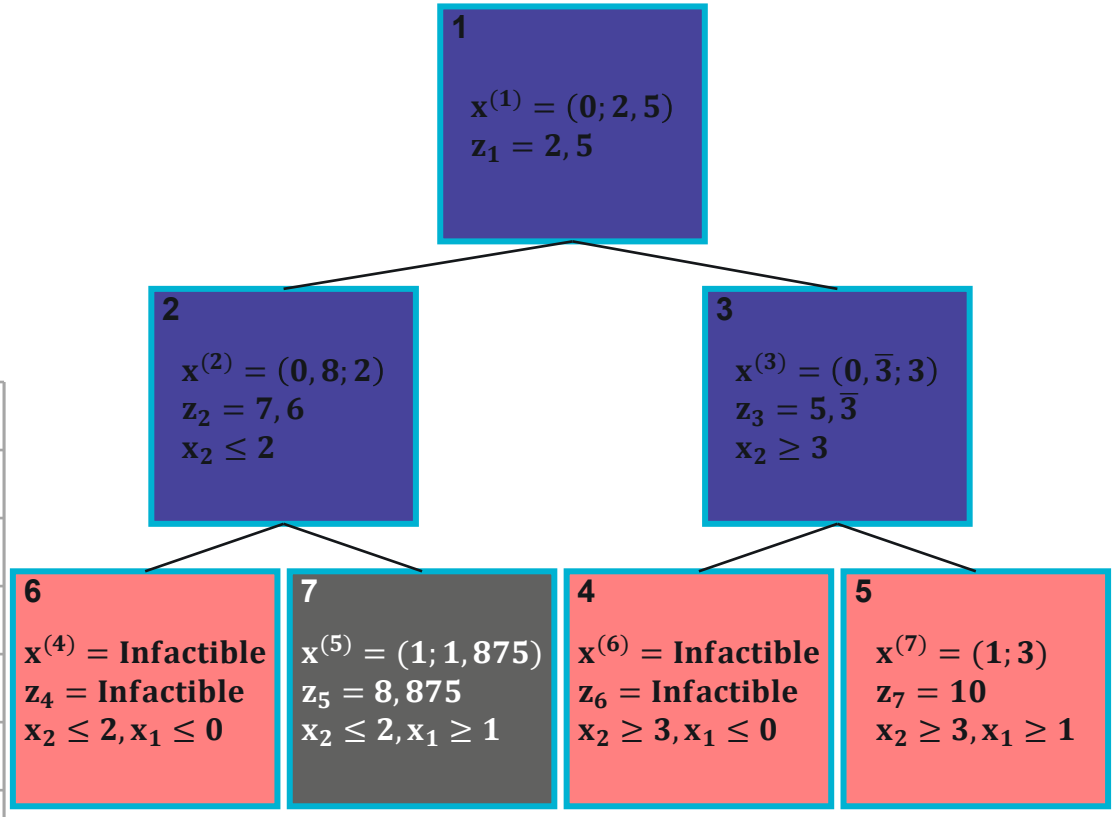


Procedamos entonces a resolver P_5

P_5

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2, x_1 \geq 1$$
$$x_1, x_2 \geq 0$$

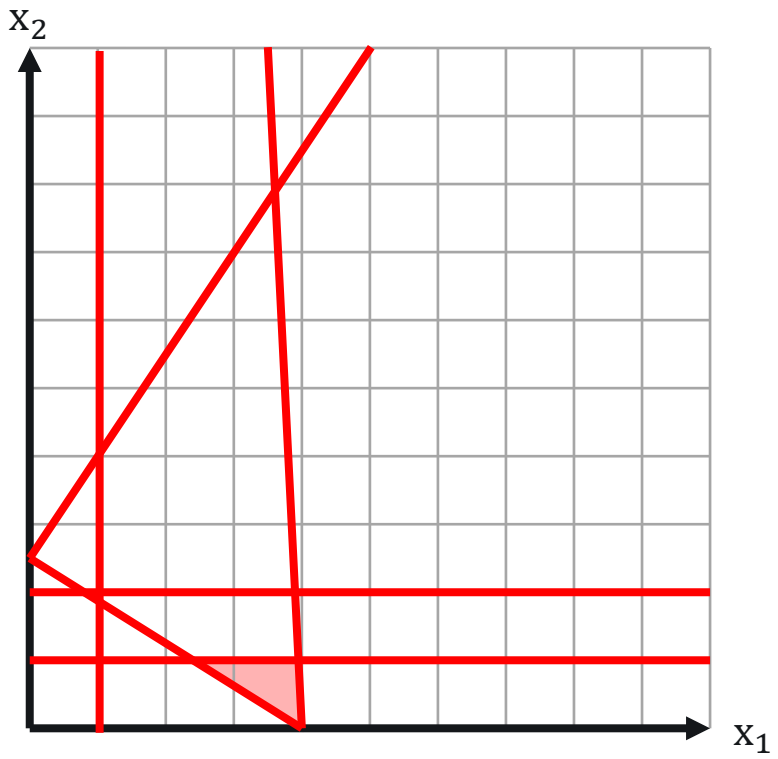




Por lo tanto, podemos ramificar P_5 en 2 sub problemas

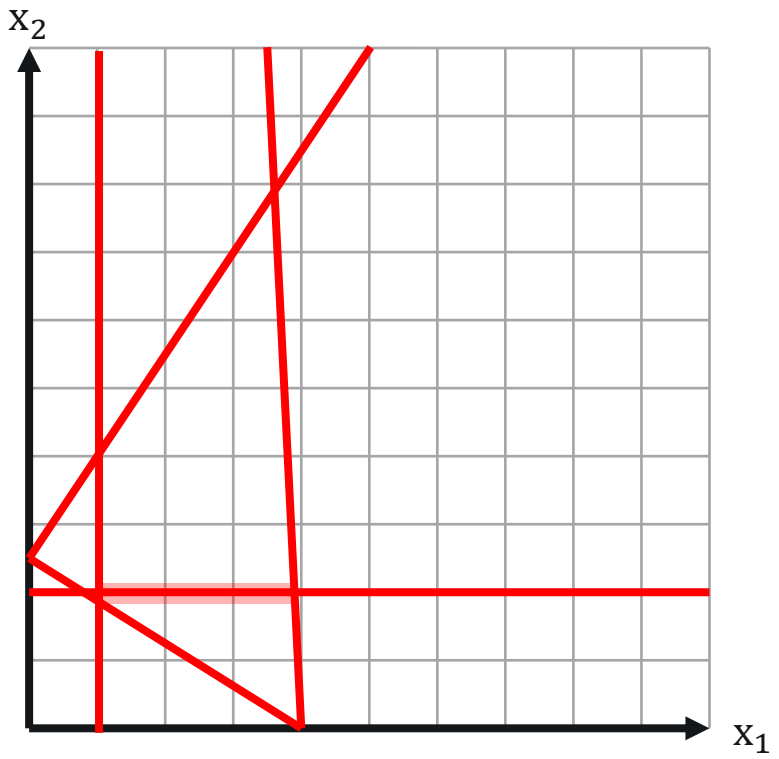
P_8

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2, x_1 \geq 1, x_2 \leq 1$$
$$x_1, x_2 \geq 0$$



P_9

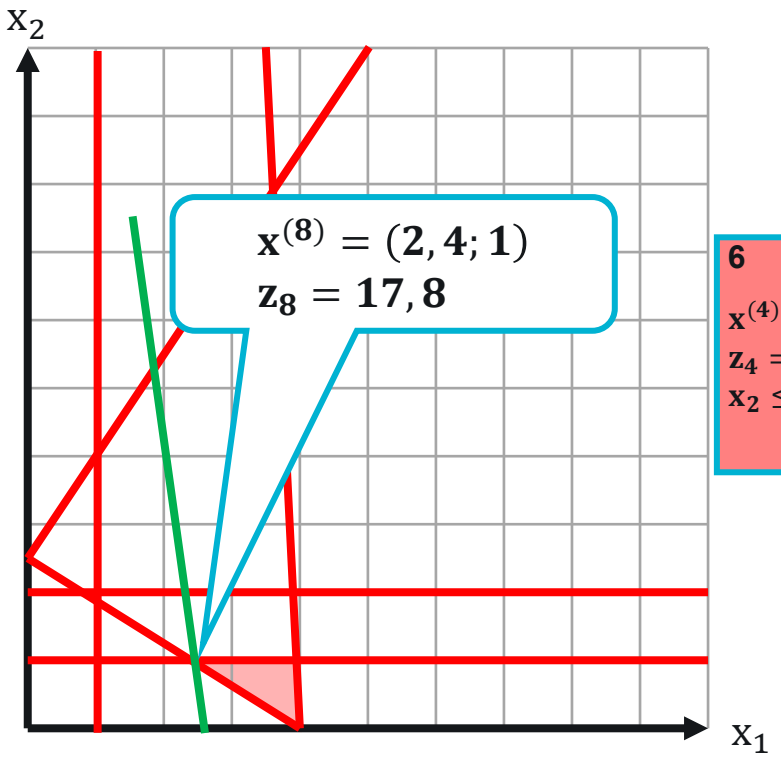
$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2, x_1 \geq 1, x_2 \geq 2$$
$$x_1, x_2 \geq 0$$

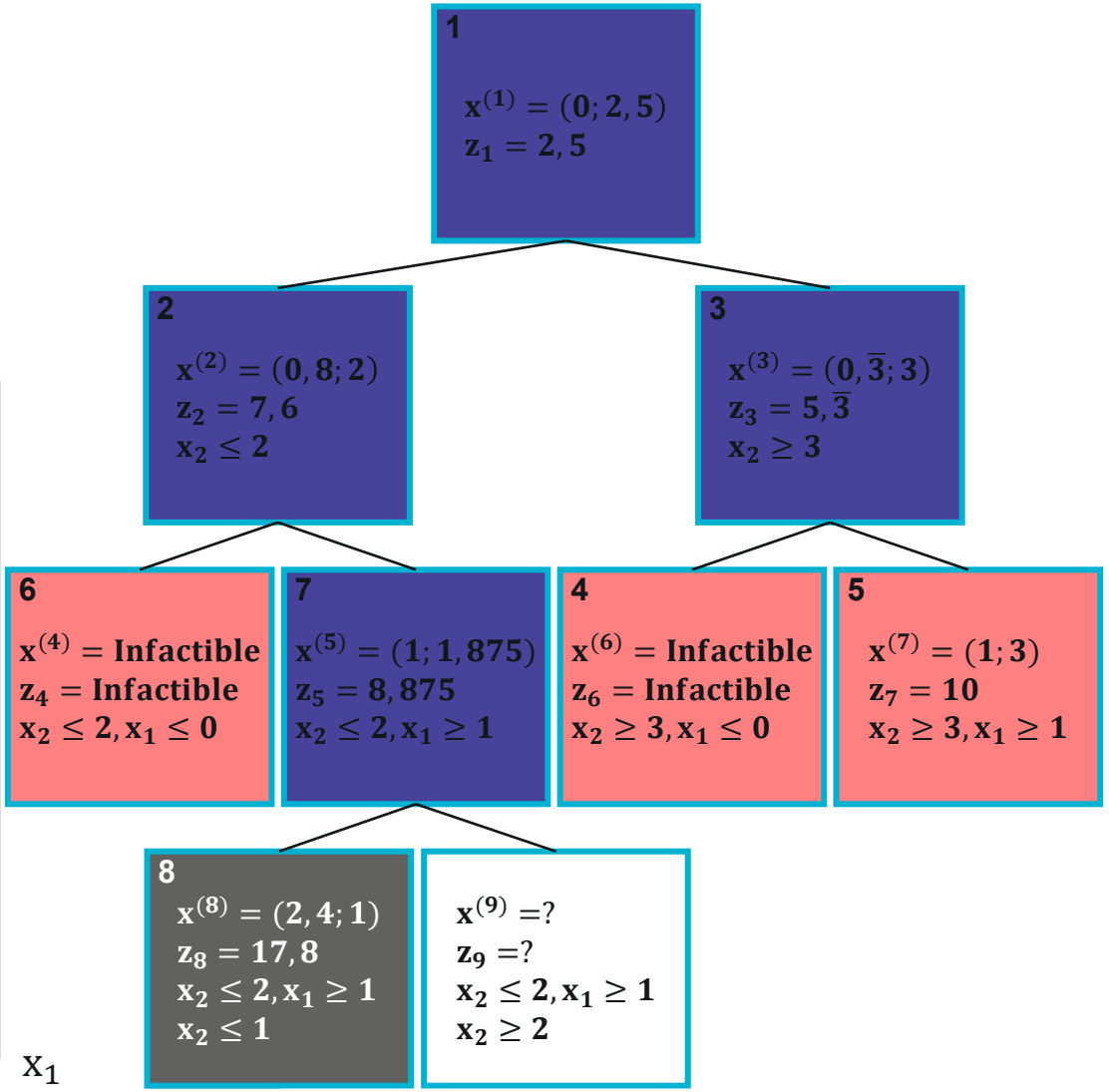


Procedamos entonces a resolver P_8, como $z_8 > I$ no ramificamos el problema

P_8

$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2, x_1 \geq 1, x_2 \leq 1$$
$$x_1, x_2 \geq 0$$

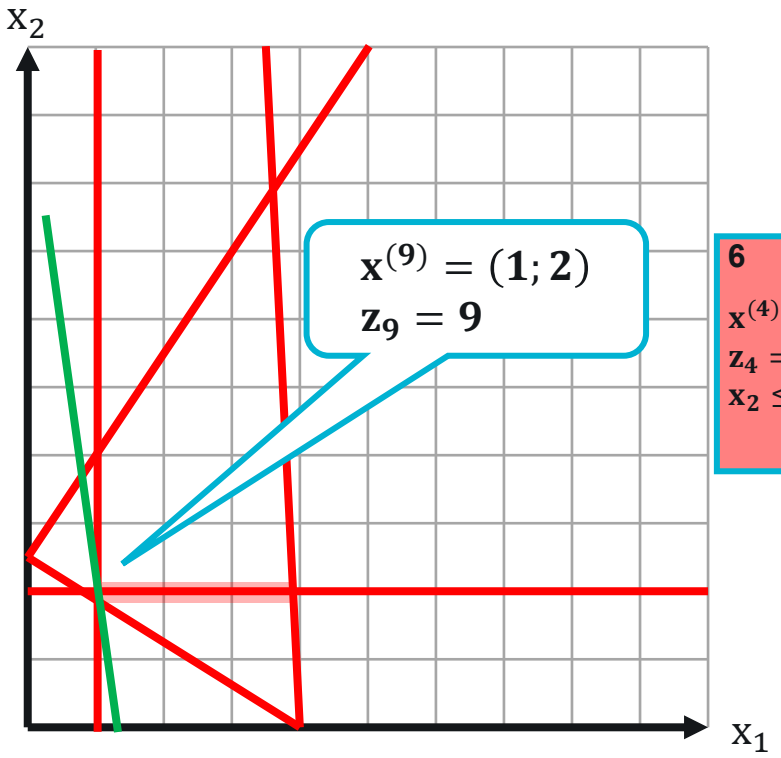


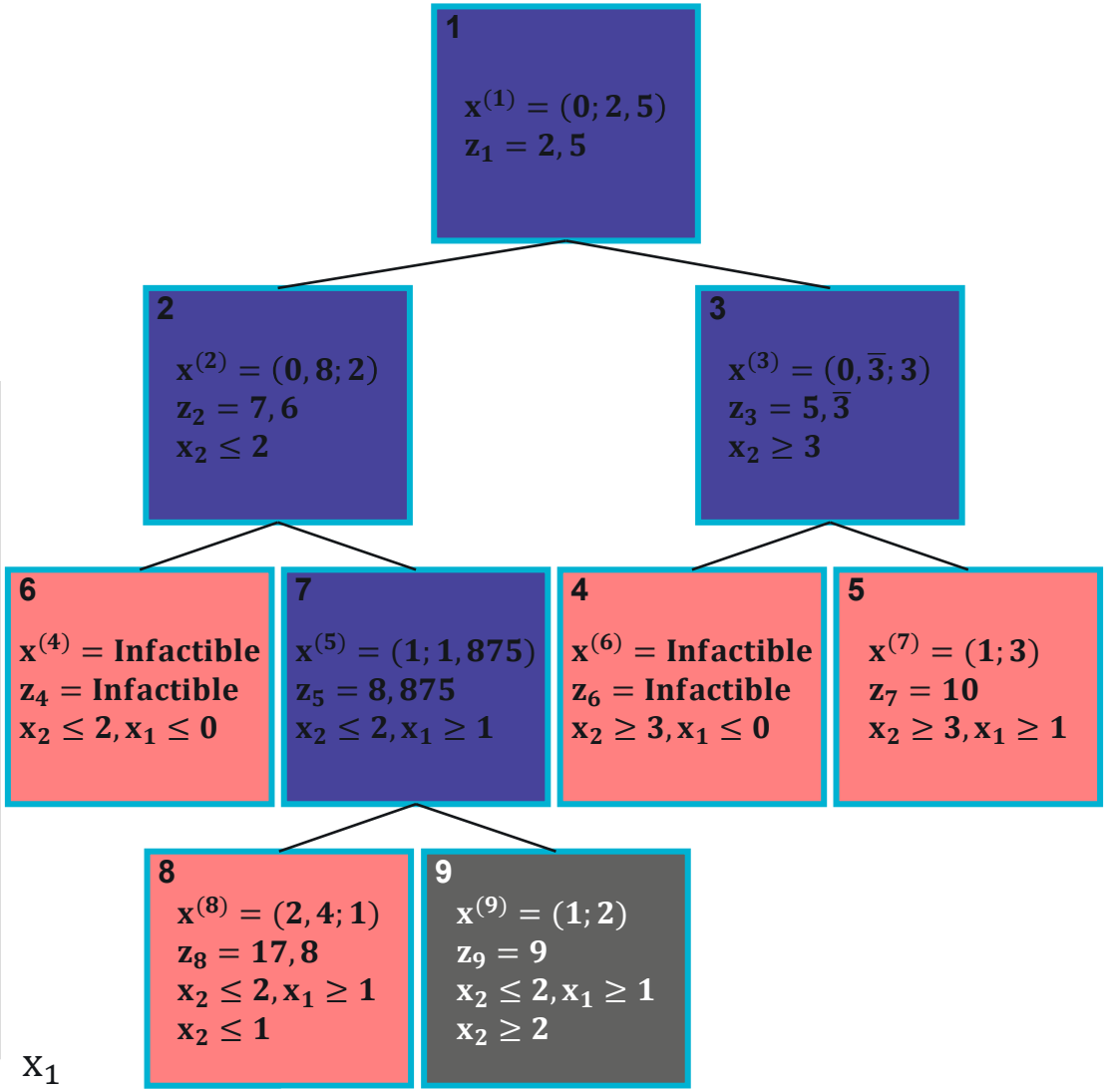


Procedamos entonces a resolver P_9, Cuyo óptimo es solución factible en el problema original, por lo que no ramificamos

P_9

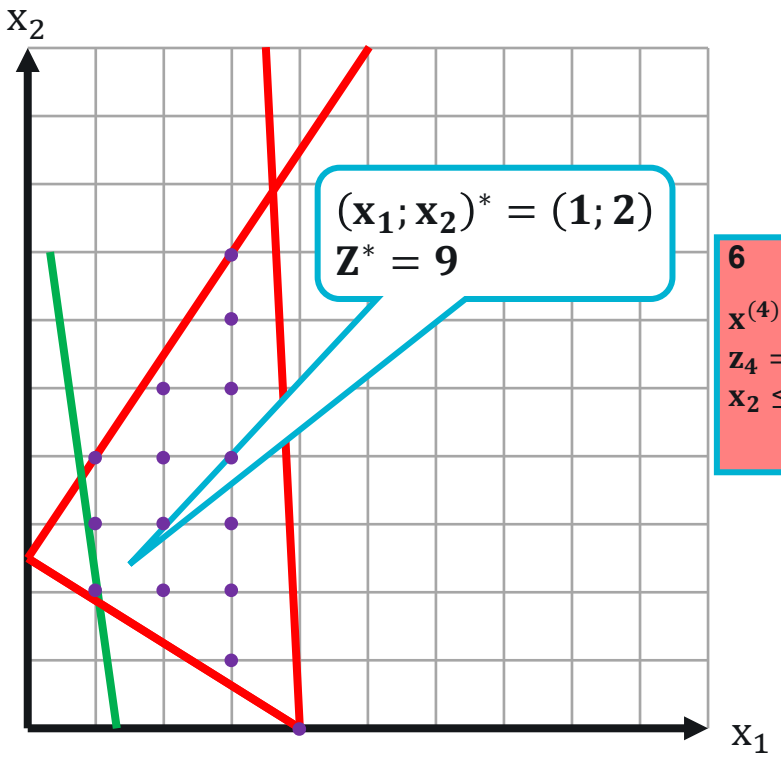
$$\text{MIN } Z = 7x_1 + x_2$$
s. a.
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_2 \leq 2, x_1 \geq 1, x_2 \geq 2$$
$$x_1, x_2 \geq 0$$

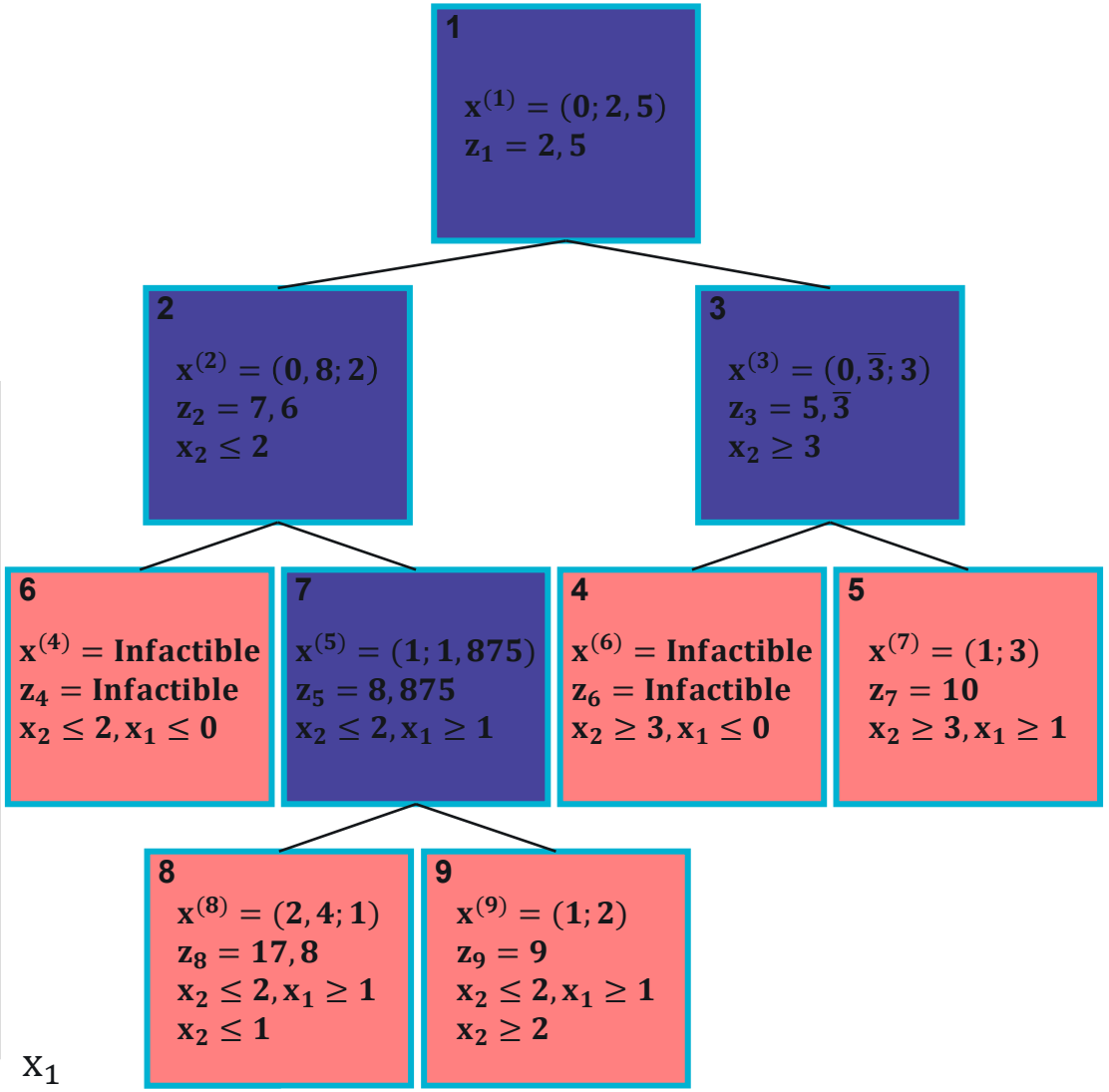




Ya no quedan más problemas que resolver por lo que el algoritmo termina y la solución óptima es $x = (1; 2)$ y $Z^* = 9$

$$\text{MIN } Z = 7x_1 + x_2$$
$$\text{s. a.}$$
$$-3x_1 + 2x_2 \leq 5$$
$$5x_1 + 8x_2 \geq 20$$
$$20x_1 + x_2 \leq 80$$
$$x_1, x_2 \in Z_0^+$$





¿Qué ocurre con el caso binario?

La ramificación no genera restricciones, sino que fija las variables en los valores 0 ó 1.

Esta situación crea problemas más simples de resolver, ya que no es necesario añadir restricciones.

Sin embargo, se requiere llevar adelante el mismo procedimiento que en el caso entero.

Ejemplo – Solo Variables binarias

Resolvamos el siguiente problema utilizando el algoritmo de Branch & Bound

$$\text{MAX } Z = 2x_1 + x_2 + 2x_3$$
$$\text{s. a.}$$
$$x_1 + 2x_2 + x_3 \leq 3$$
$$2x_1 + 2x_2 + 2x_3 \leq 3$$
$$x_1 + 2x_2 + 2x_3 \leq 4$$
$$x_1, x_2, x_3 \in \{0,1\}$$

Relajación lineal

P_1

$$\text{MAX } Z = 2x_1 + x_2 + 2x_3$$
$$\text{s. a.}$$
$$x_1 + 2x_2 + x_3 \leq 3$$
$$2x_1 + 2x_2 + 2x_3 \leq 3$$
$$x_1 + 2x_2 + 2x_3 \leq 4$$
$$0 \leq x_1, x_2, x_3 \leq 1$$

$$\mathbf{x^{(1)} = (1; 0; 0,5)}$$
$$\mathbf{z_1 = 3}$$

Por lo tanto, podemos ramificar P_1 en dos sub-problemas

P_2

$$\text{MAX } Z = 2x_1 + x_2$$
$$\text{s. a.}$$
$$x_1 + 2x_2 \leq 3$$
$$2x_1 + 2x_2 \leq 3$$
$$x_1 + 2x_2 \leq 4$$
$$0 \leq x_1, x_2 \leq 1$$
$$x_3 = 0$$

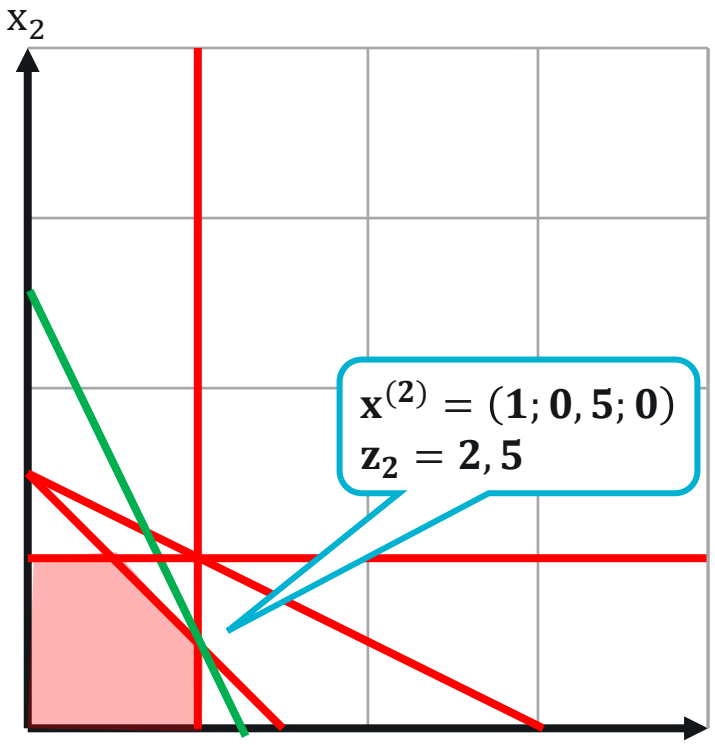
P_3

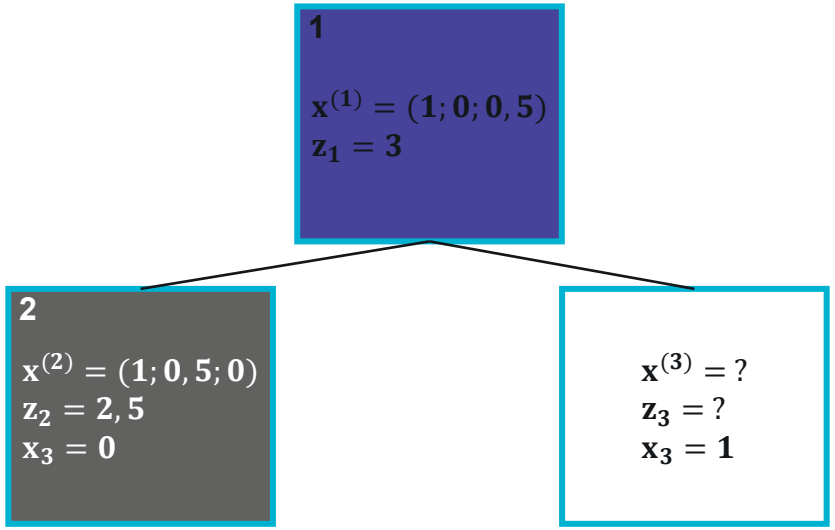
$$\text{MAX } Z = 2x_1 + x_2 + 2$$
$$\text{s. a.}$$
$$x_1 + 2x_2 \leq 2$$
$$2x_1 + 2x_2 \leq 1$$
$$x_1 + 2x_2 \leq 2$$
$$0 \leq x_1, x_2 \leq 1$$
$$x_3 = 1$$

Procedamos entonces a resolver P_2

P_2

$$\text{MAX } Z = 2x_1 + x_2$$
s. a.
$$x_1 + 2x_2 \leq 3$$
$$2x_1 + 2x_2 \leq 3$$
$$x_1 + 2x_2 \leq 4$$
$$0 \leq x_1, x_2 \leq 1$$
$$x_3 = 0$$





Por lo tanto, podemos ramificar P_2 en 2 sub problemas

P_4

$$\text{MAX } Z = 2x_1$$
$$\text{s. a.}$$
$$x_1 \leq 3$$
$$2x_1 \leq 3$$
$$x_1 \leq 4$$
$$0 \leq x_1 \leq 1$$
$$x_3 = x_2 = 0$$

P_5

$$\text{MAX } Z = 2x_1 + 1$$
$$\text{s. a.}$$
$$x_1 \leq 1$$
$$2x_1 \leq 1$$
$$x_1 \leq 2$$
$$0 \leq x_1 \leq 1$$
$$x_3 = 0, x_2 = 1$$

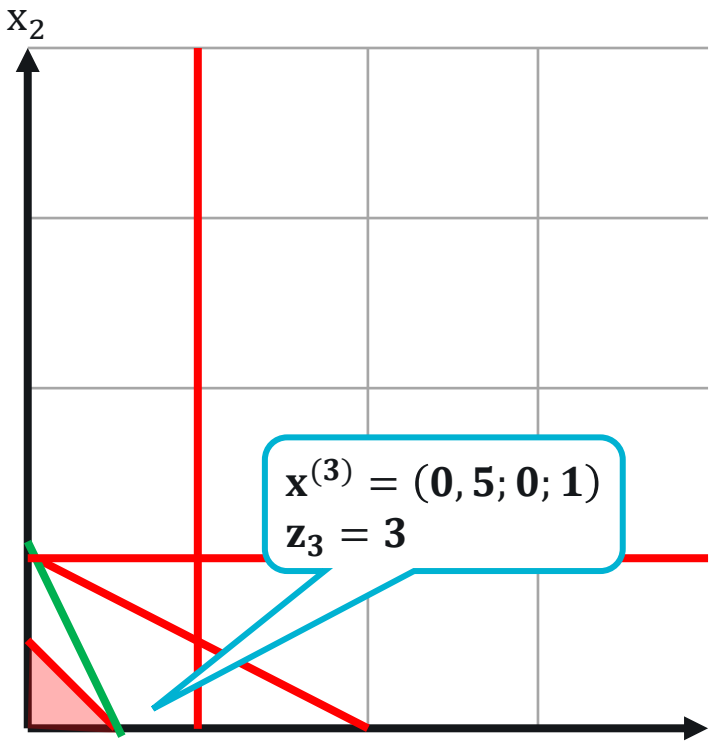
Procedamos entonces a resolver P_3

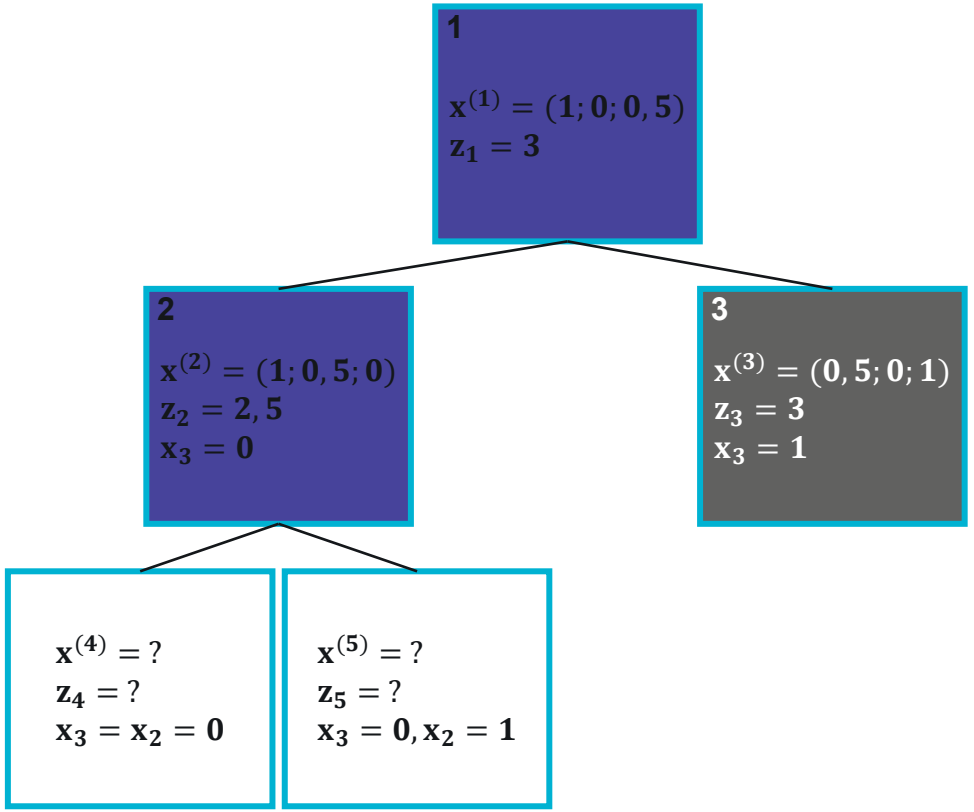
P_3

$$\text{MAX } Z = 2x_1 + x_2 + 2$$

s. a.

$$x_1 + 2x_2 \leq 2$$
$$2x_1 + 2x_2 \leq 1$$
$$x_1 + 2x_2 \leq 2$$
$$0 \leq x_1, x_2 \leq 1$$
$$x_3 = 1$$





Por lo tanto, podemos ramificar P_3 en 2 sub problemas

P_6

$$\text{MAX } Z = x_2 + 2$$
s. a.
$$2x_2 \leq 2$$
$$2x_2 \leq 1$$
$$2x_2 \leq 2$$
$$0 \leq x_2 \leq 1$$
$$x_3 = 1, x_1 = 0$$

P_7

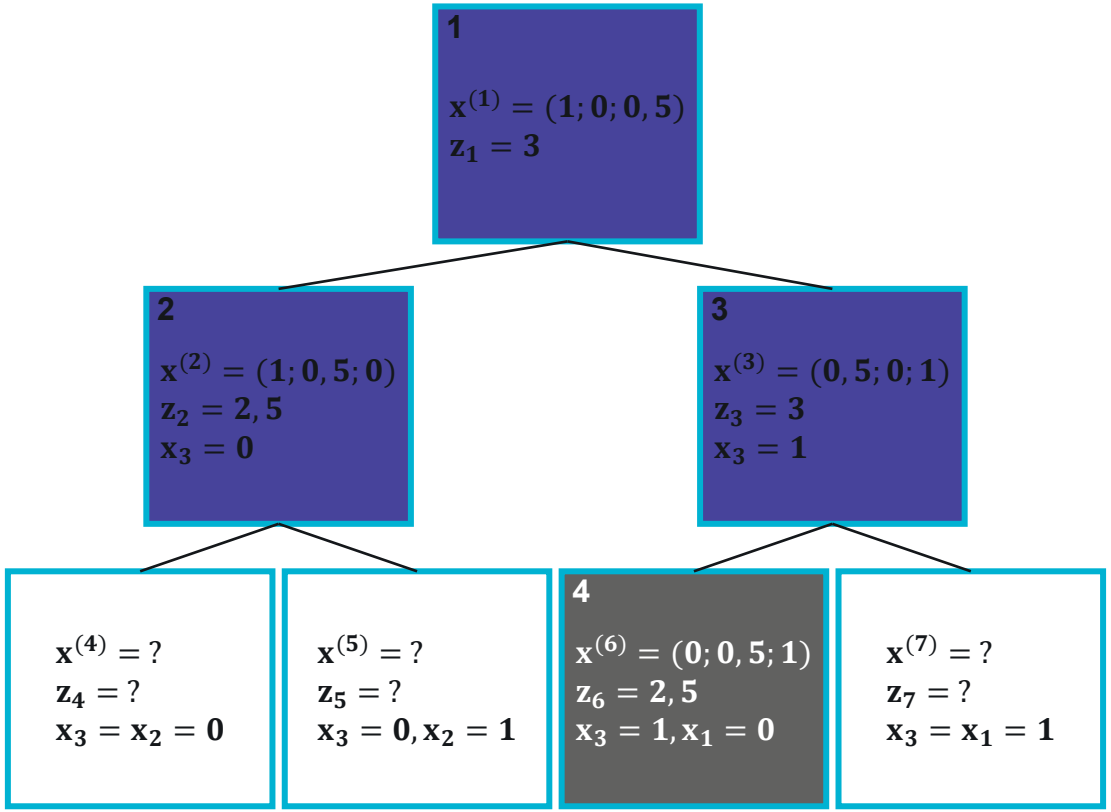
$$\text{MAX } Z = x_2 + 4$$
s. a.
$$2x_2 \leq 1$$
$$2x_2 \leq -1$$
$$2x_2 \leq 1$$
$$0 \leq x_2 \leq 1$$
$$x_3 = x_1 = 1$$

Procedamos entonces a resolver P_6

P_6

$$\text{MAX } Z = x_2 + 2$$
s. a.
$$2x_2 \leq 2$$
$$2x_2 \leq 1$$
$$2x_2 \leq 2$$
$$0 \leq x_2 \leq 1$$
$$x_3 = 1, x_1 = 0$$

$\mathbf{x^{(6)} = (0; 0,5; 1)}$
$\mathbf{z_6 = 2,5}$



Por lo tanto, podemos ramificar P_6 en 2 sub problemas

P_8

$$\text{MAX } Z = 2$$
s. a.
$$0 \leq 2$$
$$0 \leq 1$$
$$0 \leq 2$$
$$x_3 = 1, x_1 = x_2 = 0$$

P_9

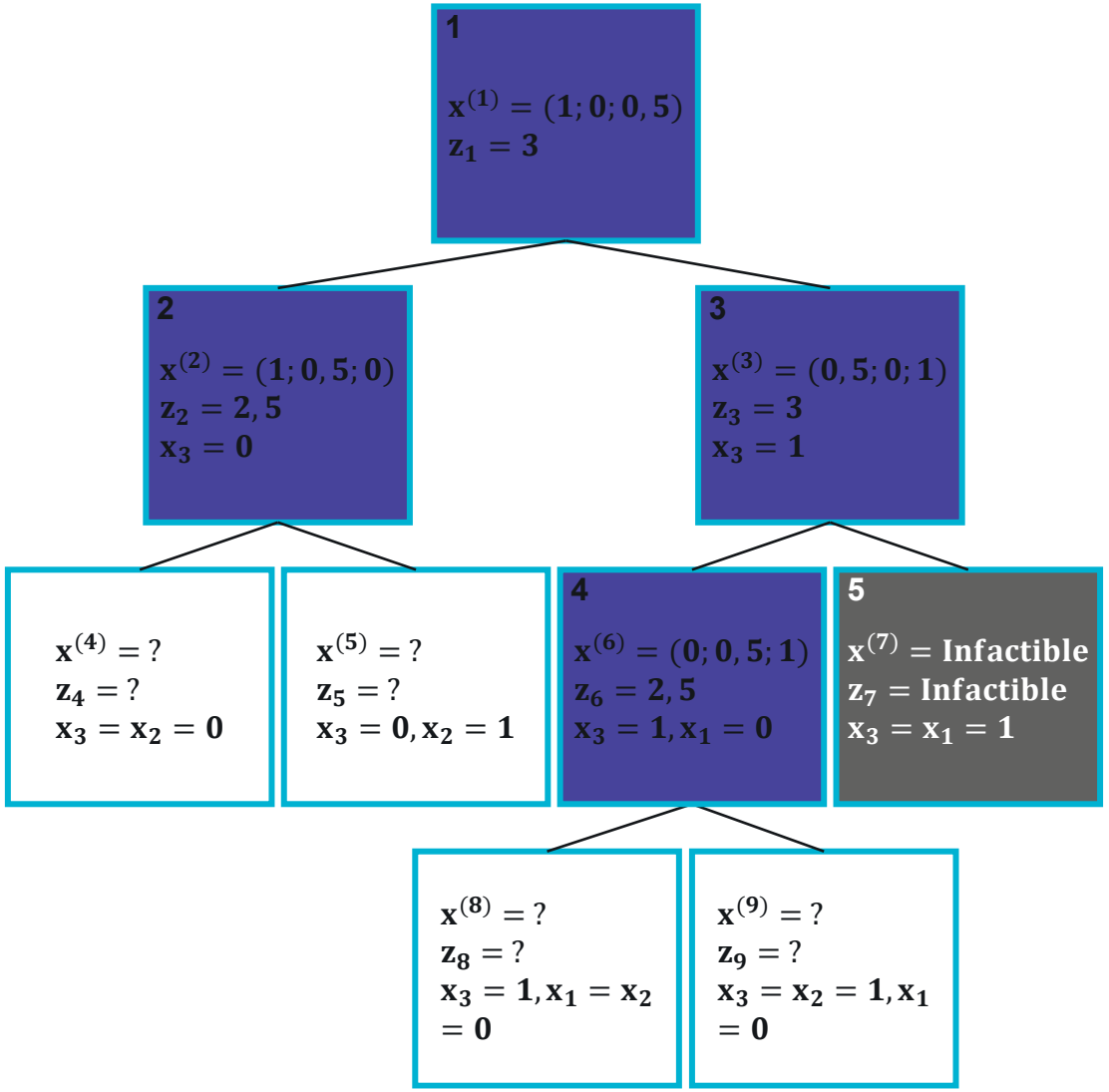
$$\text{MAX } Z = 3$$
s. a.
$$2 \leq 2$$
$$2 \leq 1$$
$$2 \leq 2$$
$$x_3 = x_2 = 1, x_1 = 0$$

Procedamos entonces a resolver P_7, el cual nos damos cuenta que es infactible por lo que no lo ramificamos

P_7

$$\text{MAX } Z = x_2 + 4$$
s. a.
$$2x_2 \leq 1$$
$$2x_2 \leq -1$$
$$2x_2 \leq 1$$
$$0 \leq x_2 \leq 1$$
$$x_3 = x_1 = 1$$

Infactible

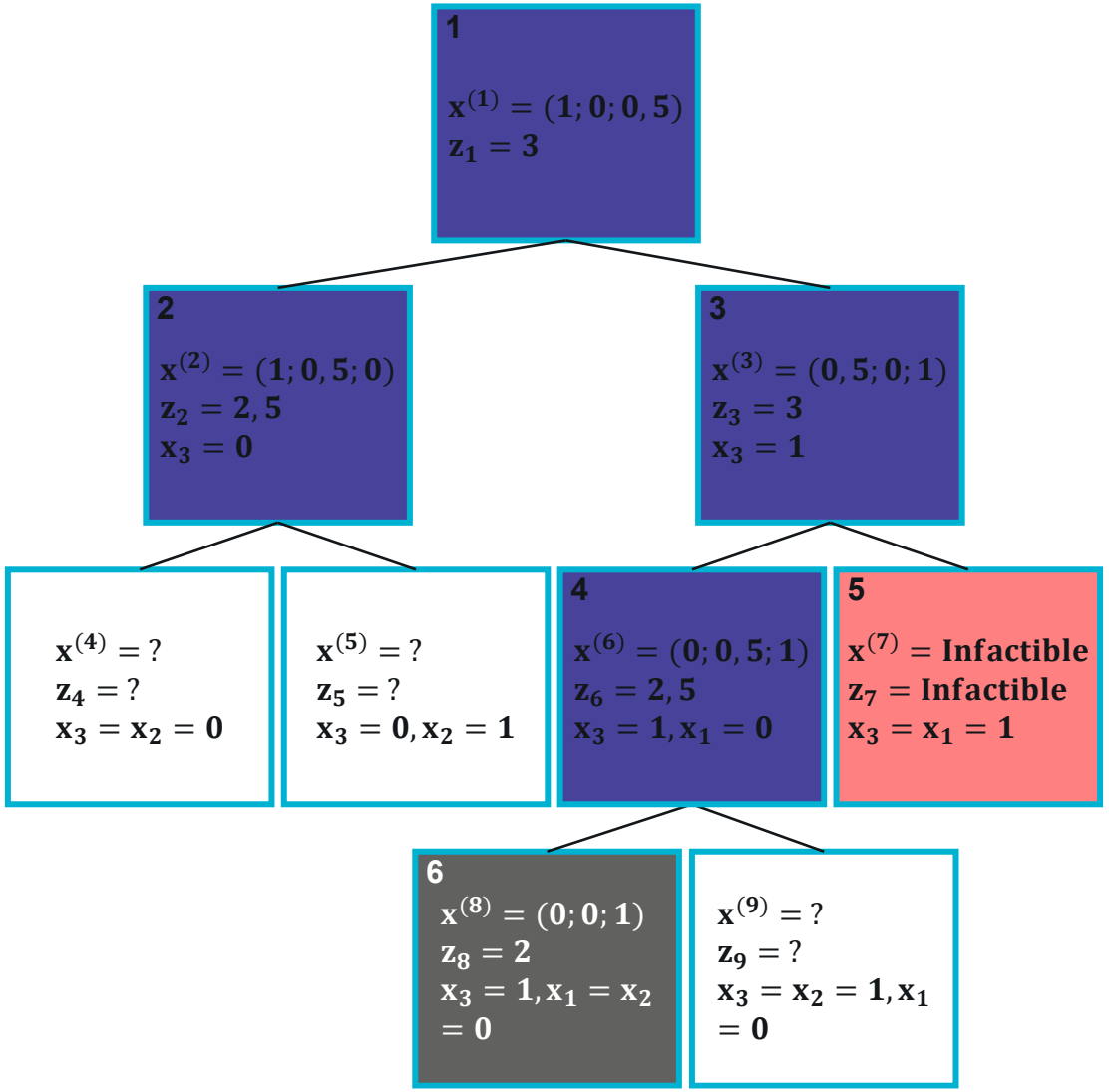


Procedamos entonces a resolver P_9, Cuyo óptimo es solución factible en el problema original, por lo que no ramificamos

P_8

$\text{MAX } Z = 2$
s. a.
$0 \leq 2$
$0 \leq 1$
$0 \leq 2$
$x_3 = 1, x_1 = x_2 = 0$

$\mathbf{x^{(8)} = (0; 0; 1)}$
$\mathbf{z_8 = 2}$

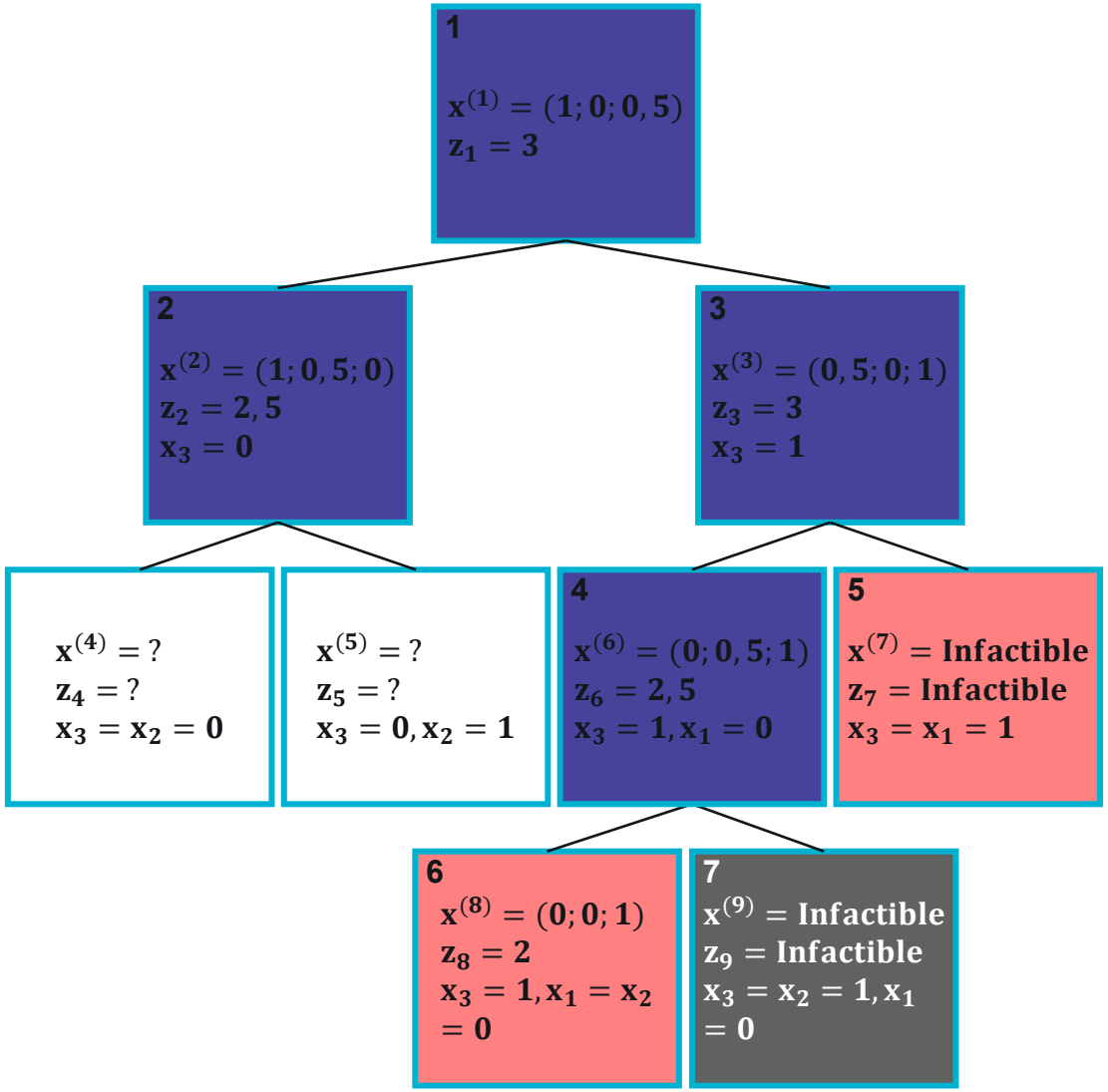


Procedamos entonces a resolver P_8, el cual nos damos cuenta que es infactible por lo que no lo ramificamos

P_9

$$\text{MAX } Z = 3$$
s. a.
$$2 \leq 2$$
$$2 \leq 1$$
$$2 \leq 2$$
$$x_3 = x_2 = 1, x_1 = 0$$

Infactible

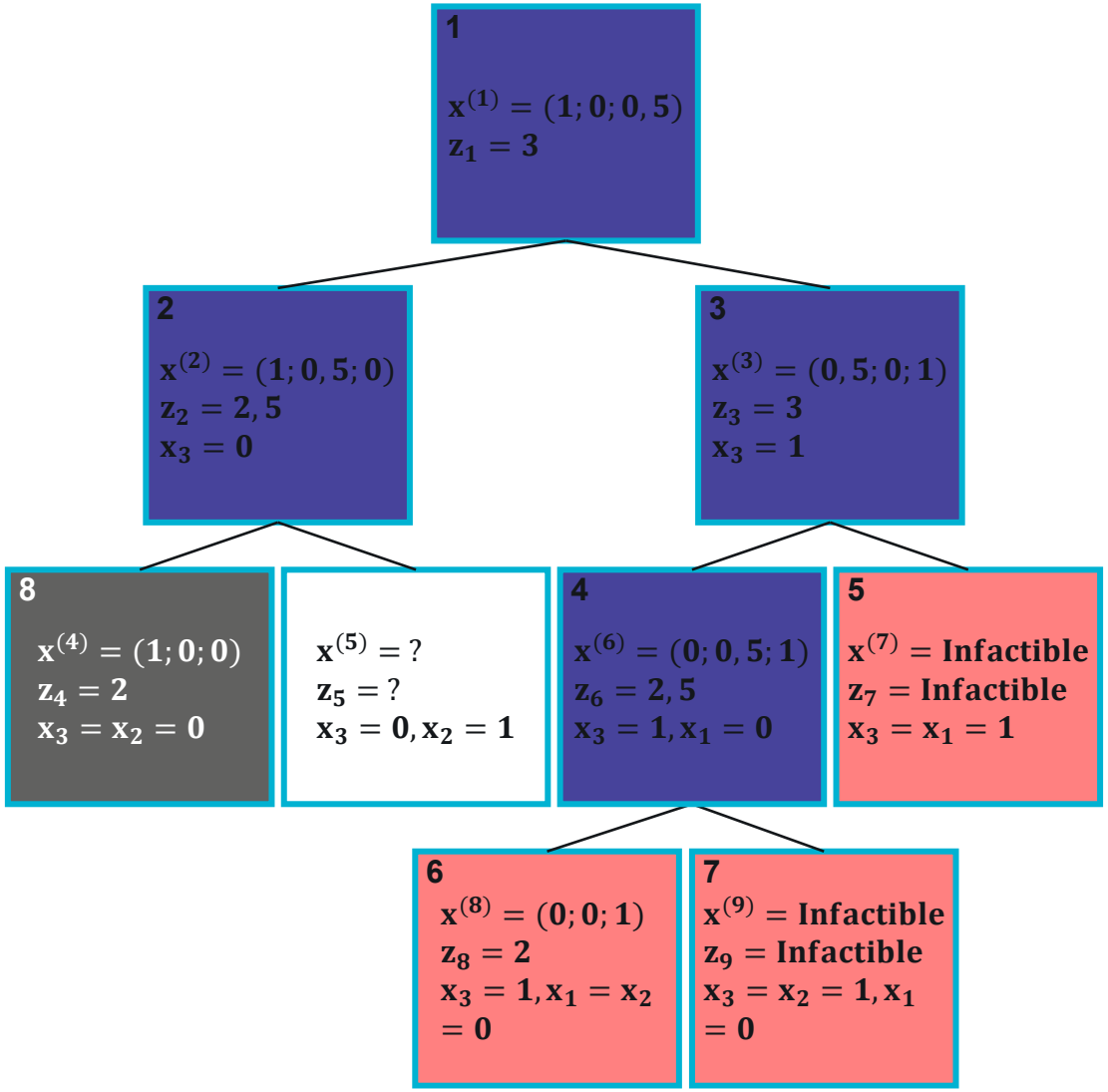


Procedamos entonces a resolver P_4, Cuyo óptimo es solución factible en el problema original, por lo que no ramificamos

P_4

$$\text{MAX } Z = 2x_1$$
s. a.
$$x_1 \leq 3$$
$$2x_1 \leq 3$$
$$x_1 \leq 4$$
$$0 \leq x_1 \leq 1$$
$$x_3 = x_2 = 0$$

$\mathbf{x^{(4)} = (1; 0; 0)}$
$\mathbf{z_4 = 2}$

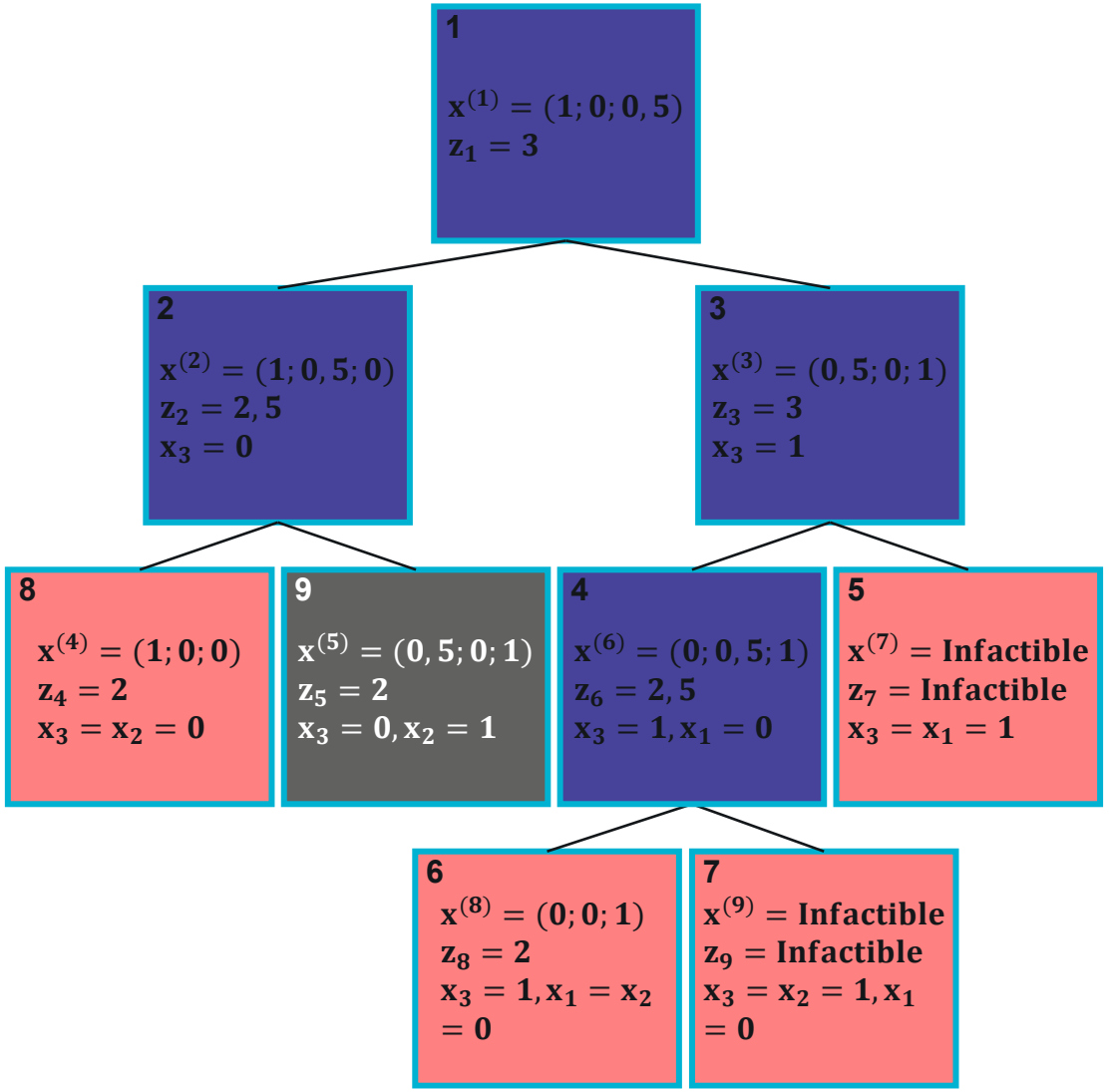


Procedamos entonces a resolver P_5, como $z_5 \leq I$ no ramificamos el problema

P_5

$$\text{MAX } Z = 2x_1 + 1$$
s. a.
$$x_1 \leq 1$$
$$2x_1 \leq 1$$
$$x_1 \leq 2$$
$$0 \leq x_1 \leq 1$$
$$x_3 = 0, x_2 = 1$$

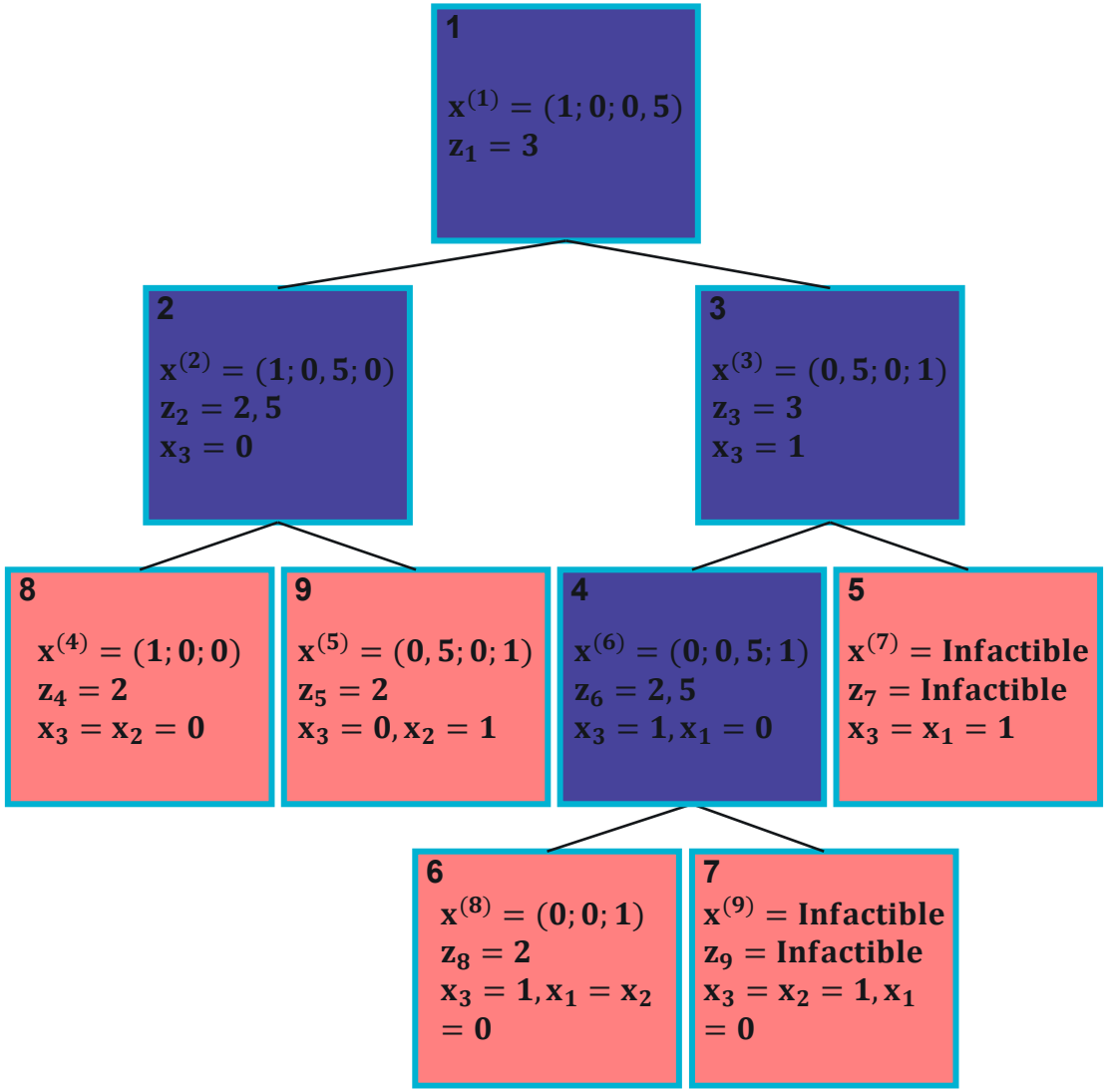
$\mathbf{x^{(5)} = (0,5; 0; 1)}$
$\mathbf{z_5 = 2}$



Ya no quedan más problemas que resolver. el algoritmo termina y la solución óptima es $x = (0; 0; 1)$ o $x = (1; 0; 0)$ y $Z^* = 2$

$$\text{MAX } Z = 2x_1 + x_2 + 2x_3$$
s. a.
$$x_1 + 2x_2 + x_3 \leq 3$$
$$2x_1 + 2x_2 + 2x_3 \leq 3$$
$$x_1 + 2x_2 + 2x_3 \leq 4$$
$$x_1, x_2, x_3 \in \{0,1\}$$

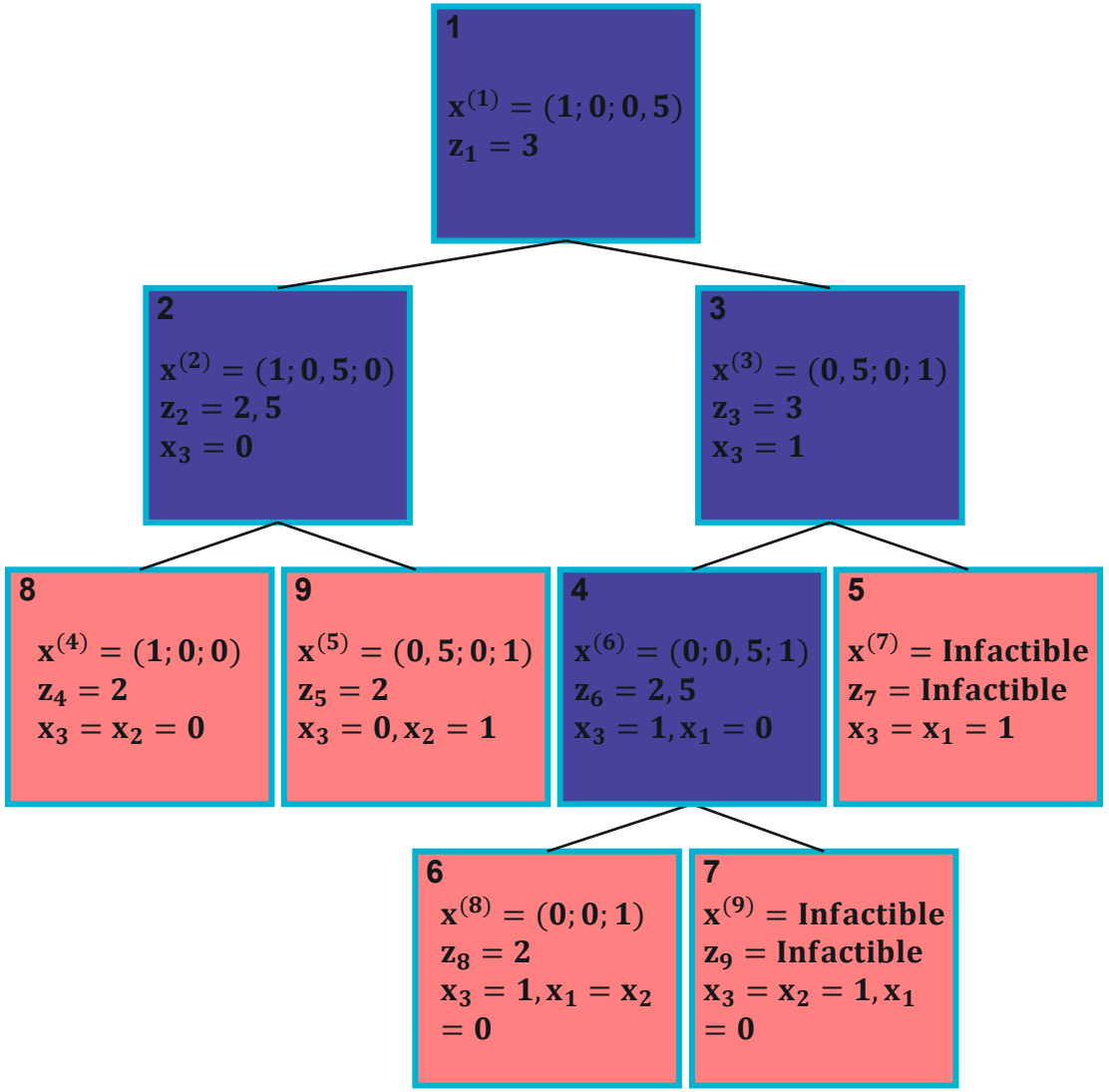
$(x_1; x_2: x_3)^* = (0; 0; 1)$ y $(1; 0; 0)$
$Z^* = 2$



Ya no quedan más problemas que resolver. el algoritmo termina y la solución óptima es $x = (0; 0; 1)$ o $x = (1; 0; 0)$ y $Z^* = 2$

$$\text{MAX } Z = 2x_1 + x_2 + 2x_3$$
s. a.
$$x_1 + 2x_2 + x_3 \leq 3$$
$$2x_1 + 2x_2 + 2x_3 \leq 3$$
$$x_1 + 2x_2 + 2x_3 \leq 4$$
$$x_1, x_2, x_3 \in \{0,1\}$$

$(x_1; x_2: x_3)^* = (0; 0; 1)$ y $(1; 0; 0)$
$Z^* = 2$



UNIVERSIDAD ADOLFO IBÁÑEZ

FACULTAD DE INGENIERÍA Y CIENCIAS